

Quantum Advantage for Hidden Subgroup Problems

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The Hidden Subgroup Problem

The Hidden Subgroup Problem (HSP) is a group theoretic problem whose solution has become fundamental in quantum computation. Every known exponential speed-up of classical computers via quantum computers, such as Shor's factoring algorithm and the discrete log algorithm, are instances of the HSP over finite abelian groups.

Consider a black box function $f : G \rightarrow S$ where G is a finite abelian group and S is a finite set. Let H be some unknown subgroup of G . A function that *hides* H satisfies:

$$\begin{aligned} f(x) = f(y) &\iff y - x \in H \\ &\iff y = x + h \text{ for some } h \in H \end{aligned}$$

or equivalently that f is constant on cosets of H in G and distinct on different cosets. The **Hidden Subgroup Problem** is to (efficiently) determine H or a generating set for H using queries to f .

In the following discussion of computational approaches to the HSP, we will encode each element of G as a computational basis state using $n = \lceil \log |G| \rceil$ qubits. Then the size of G is exponential in n . In the classical computational approach, in order to recover any information about H we need to find $x, y \in G$ such that x, y are in the same coset of H . This requires $\Omega(\sqrt{|G|}) = \Omega(2^{n/2})$ queries to f .

Characters of a Group

In order to solve the HSP, we notice that for any $h \in H$, $f(x+h) = f(x)$, $\forall x \in G$ inducing a periodicity on f . To study this further we introduce the characters of a group, a group theoretic analogue to frequencies in the classical sense.

A **character** of a finite abelian group is group homomorphism $\chi : G \rightarrow U(1)$. The set of all characters of G forms another group $\hat{G}$ under multiplication. For finite abelian Groups we have $G \cong \hat{\hat{G}}$.

In the case of a cyclic group, isomorphic to $\mathbb{Z}_n$, each $\chi \in \hat{G}$ is determined entirely based on where it maps the generator element g . Further since $ng = 0$, and any homomorphism must map the group identity 0 to 1, $\chi(g)$ must be an n -th root of unity. We can then define $\chi_y(x) = e^{2\pi i xy/n}$ for $y \in \mathbb{Z}_n$ to be the unique element of $\hat{G}$ mapping the generator to $e^{2\pi i y/n}$.

For an arbitrary finite abelian group, recall that $G \cong \mathbb{Z}_{n_1} \times \dots \times \mathbb{Z}_{n_k}$. Since homomorphisms on G must map the generator of each factor $\mathbb{Z}_{n_i}$ to an n_i th root of unity, the characters of G are defined as follows: for $y \in \mathbb{Z}_{n_1} \times \dots \times \mathbb{Z}_{n_k}$, the corresponding character evaluated at x is $\chi_y(x) = \prod_{i=1}^k e^{2\pi i x_i y_i / n_i}$. With these definitions we can derive the Quantum Fourier Transform, an essential subroutine for solving the HSP.

Quantum Fourier Transform

We show that the quantum Fourier transform can be constructed in a polynomial number of gates, playing a pivotal role in algorithms for the HSP that provide an exponential speed-up. The QFT can be written as:

$$F_G := \frac{1}{\sqrt{|G|}} \sum_{x \in G} \sum_{y \in \hat{G}} \chi_y(x) |y\rangle \langle x| \quad (1)$$

To construct the circuit for the QFT, we can derive an expression for the QFT in *product-state* form. Consider the case where $G = \mathbb{Z}_{2^n}$. Then for any basis state $|j\rangle$:

$$|j\rangle \rightarrow \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} e^{2\pi i j k / 2^n} |k\rangle \quad (2)$$

$$= \frac{1}{2^{n/2}} \sum_{k_1=0}^1 \dots \sum_{k_n=0}^1 e^{2\pi i j \left(\sum_{l=1}^n k_l 2^{-l} \right)} |k_1 \dots k_n\rangle \quad (3)$$

$$= \frac{1}{2^{n/2}} \sum_{k_1=0}^1 \dots \sum_{k_n=0}^1 \bigotimes_{l=1}^n e^{2\pi i j k_l 2^{-l}} |k_l\rangle \quad (4)$$

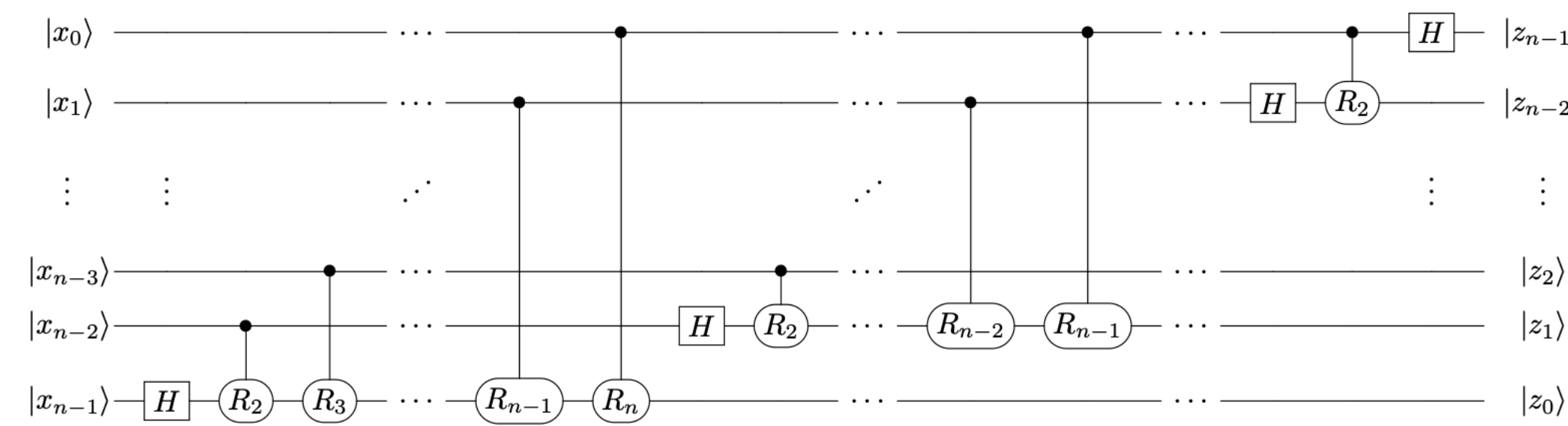
$$= \frac{1}{2^{n/2}} \bigotimes_{l=1}^n \left[\sum_{k_l=0}^1 e^{2\pi i j k_l 2^{-l}} |k_l\rangle \right] \quad (5)$$

$$= \frac{(|0\rangle + e^{2\pi i 0 \cdot j n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot j n - 1 j n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot j 1 2^{-1} \dots j n} |1\rangle)}{2^{n/2}} \quad (6)$$

Quantum Fourier Transform, continued

Therefore letting $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ and $R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i / 2^k} \end{bmatrix}$ we get the following circuit

using $O(t^2)$ gates for a t qubit system:



After this computation, we need at most $t/2$ swap gates to restore the correct order of the qubits, thus giving an overall complexity of $O(t^2)$ gates.

Phase Estimation and QFT over $\mathbb{Z}_n$

We then construct the quantum Fourier transform over an arbitrary cyclic group $\mathbb{Z}_n$. More specifically, we want to take $|x, 0\rangle \rightarrow |0, \tilde{x}\rangle$ where $F_{\mathbb{Z}_n} |x\rangle = |\tilde{x}\rangle$.

First, the mapping $|x, 0\rangle \mapsto |x, \tilde{x}\rangle$ can be implemented using quadratically many gates. Next, we make use of the quantum phase estimation (QPE) subroutine, which uses our previous construction of the QFT over $\mathbb{Z}_{2^n}$. Given an eigenstate $|u\rangle$ with eigenvalue $e^{2\pi i \phi}$ of a unitary U , QPE writes a t -bit estimate of the phase ϕ into a control register, mapping $|0, u\rangle \mapsto |\phi, u\rangle$. Then, consider the unitary $U := \sum_{x \in \mathbb{Z}_n} |x+1\rangle \langle x|$, which has eigenstates $|\tilde{x}\rangle$ with eigenvalue $e^{-2\pi i x/n}$. Therefore the quantum phase estimation algorithm can take $|\tilde{x}, 0\rangle \mapsto |\tilde{x}, x\rangle$. We can then use the inverse of quantum phase estimation to do this process in reverse, thus allowing us to take:

$$|x, 0\rangle \rightarrow |x, \tilde{x}\rangle \xrightarrow{QPE^\dagger} |0, \tilde{x}\rangle$$

Using the Fundamental Theorem of Finite Abelian Groups, we can decompose any abelian group G into the direct product of cyclic groups of prime power order. Since the QFT over the direct product of groups is the tensor product of QFTs over the individual groups, we can use our QFT over $\mathbb{Z}_n$, to solve the hidden subgroup problem over arbitrary finite abelian groups.

An Algorithm for the Abelian HSP

Using our construction of the quantum Fourier transform over $\mathbb{Z}_n$ and given a black-box that implements unitary $U |g, h\rangle \rightarrow |g, h \oplus f(g)\rangle$, we can perform the following efficient quantum algorithm to solve the hidden subgroup problem over finite abelian group G :

1. We begin by creating a superposition of elements in G and then computing the function value in another register:

$$\frac{1}{\sqrt{|G|}} \sum_{x \in G} |x, 0\rangle \mapsto \frac{1}{\sqrt{|G|}} \sum_{x \in G} |x, f(x)\rangle$$

2. Measure and discard the second register, collapsing the first to a uniform superposition of a coset. The post-measurement state is:

$$|x + H\rangle = \frac{1}{\sqrt{|H|}} \sum_{h \in H} |x + h\rangle$$

3. Perform a Quantum Fourier transform over G . We see that:

$$\begin{aligned} F_G |x + H\rangle &= \frac{1}{\sqrt{|H||G|}} \sum_{y \in \hat{G}} \sum_{h \in H} \chi_y(x + h) |y\rangle \\ &= \sqrt{\frac{|H|}{|G|}} \sum_{y \in \hat{G}} \chi_y(x) \chi_y(H) |y\rangle \end{aligned}$$

where we define $\chi_y(H) := \frac{1}{|H|} \sum_{h \in H} \chi_y(h)$.

An Algorithm for the Abelian HSP, continued

We can see two things: firstly, that if $\chi_y(h) = 1$ for all $h \in H$, then clearly $\chi_y(H) = 1$. Secondly if there is $h' \in H$ such that $\chi_y(h') \neq 1$ then since $h' + H = H$:

$$\begin{aligned} \chi_y(H) &= \frac{1}{|H|} \sum_{h \in h' + H} \chi_y(h) \\ &= \frac{1}{|H|} \sum_{h \in H} \chi_y(h' + h) = \frac{1}{|H|} \sum_{h \in H} \chi_y(h') \chi_y(h) \\ &= \chi_y(h') \chi_y(H) \\ &\implies \chi_y(H) = 0 \end{aligned}$$

Therefore only the terms of $y \in \hat{G}$ where all $\chi_y(h) = 1$ for all $h \in H$ remain.

$$F_G |x + H\rangle = \sqrt{\frac{|H|}{|G|}} \sum_{y: \chi_y(H)=1} \chi_y(x) |y\rangle$$

4. Measure the register and receive y_0 . We then can determine some character χ_{y_0} where for all $h \in H$, $\chi_{y_0}(h) = 1$.

$$H \subset \ker(\chi_{y_0}) = \{g \in G \mid \chi_{y_0}(g) = 1\}$$

By measuring several $y_0, \dots, y_{m-1}$, we can take the intersection of the various kernels and very quickly determine H . Define $K = \ker(\chi_{y_0}) \cap \dots \cap \ker(\chi_{y_{m-1}})$. After $m = O(\log |G|)$ samples, it is extremely likely that $K = H$. Then, for $n = \lceil \log |G| \rceil$ qubits, since each sample requires n Hadamard gates, one query of U (assume U runs in $\text{poly}(n)$ gates), and one Fourier Transform ($O(n^2)$ gates), solving the Hidden Subgroup Problem requires $O(m(\text{poly}(n))) = O(\text{poly}(n))$ gates and $O(m)$ queries. In comparison, if group G has N candidate subgroups whose only intersection is the identity element, a deterministic classical computer must make $\Omega(\sqrt{N})$ queries. For example, in Simon's problem, the hidden subgroup has form $\{0, s\}$, where $s \in \mathbb{Z}_N^n$. We can see that there are $N = 2^n - 1$ possible subgroups and thus that a deterministic classical computer would require $\Omega(2^{n/2})$ queries, demonstrating an exponential quantum advantage.

Abelian Hidden Subgroup Problems

As an example, we discuss the discrete logarithm in $\mathbb{Z}_n^\times$, the basis of the Diffie-Hellman key exchange protocol, as an instance of the hidden subgroup problem in $\mathbb{Z}_n^2$. Given element $x \in G$ where G is a cyclic group generated by g , the *discrete logarithm* of x is the smallest integer α such that $x = \alpha g$. If we define function $f(u, v) = ux + vg = (\alpha u + v)g$, f is constant on the lattice defined as:

$$\mathcal{L}_\lambda := \{(u, v) \in \mathbb{Z}_N^2 : \alpha u + v = \lambda\}$$

and thus hides subgroup:

$$\mathcal{L}_0 = \{(0, 0), (1, -\alpha), (2, -2\alpha), \dots, (N-1, -(N-1)\alpha)\}$$

Then, we can utilize the quantum algorithm described above to efficiently calculate discrete logarithms in $\mathbb{Z}_p^\times$. The table below summarizes other instances of the abelian HSP, detailing the group G , the hidden subgroup H , and the hiding function f .

	G	H	f
Simon's Problem	$\mathbb{Z}_2^n \oplus$	$\{0, s\}, s \in \mathbb{Z}_2^n$	$f(x \oplus s) = f(x)$
Period-finding	$\mathbb{Z}_r +$	$\{0, r, 2r, \dots\}$	$f(x + r) = f(x)$
Order-finding	$\mathbb{Z}_r +$	$\{0, r, 2r, \dots\}$	$f(x) = a^x, f(x + r) = f(x)$
Discrete Logarithm	$\mathbb{Z}_n^2 +$	$\mathcal{L}_0$	$f(u, v) = xu + vg, x = \alpha g$

References

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